Harin Lee

leeharin@cs.washington.edu
harinboy.github.io

Academic Interests

My research interests lie in the broad area of decision-making theory, particularly in bandits and reinforcement learning theory.

Education

University of Washington, Seattle, WA, USA — Sep 2025-Present
Advisor : Prof. Kevin Jamieson
Ph.D. Program
Paul G. Allen School of Computer Science and Engineering

Seoul National University, Seoul, Korea — Mar 2018–Feb 2025
Bachelor of Science
Double Major in Computer Science and Engineering and Mathematical Sciences
GPA : Overall 4.18/4.3, CSE 4.2/4.3, Math 4.3/4.3, 1st in department
Leave of Absence for Military Service: Sep 2020–Aug 2022

Seoul Science High School, Seoul, Korea — Mar 2015–Feb 2018

Publications

[1] **H. Lee** and K. Jamieson, "Minimax optimal strategy for delayed observations in online reinforcement learning", in *Forty-third International Conference on Machine Learning*, 2026, **(Oral)**.

[2] **H. Lee** and M.-h. Oh, "Unified framework of distributional regret in multi-armed bandits and reinforcement learning", in *The Thirty Ninth Annual Conference on Learning Theory*, 2026, (To appear).

[3] **H. Lee** and M.-h. Oh, "Infrequent exploration in linear bandits", in *Advances in Neural Information Processing Systems*, 2025.

[4] **H. Lee** and M.-h. Oh, "Minimax optimal reinforcement learning with quasi-optimism", in *The Thirteenth International Conference on Learning Representations*, 2025.

[5] **H. Lee**, T. Hwang, and M.-h. Oh, "Lasso bandit with compatibility condition on optimal arm", in *The Thirteenth International Conference on Learning Representations*, 2025.

[6] **H. Lee** and M.-h. Oh, "Improved regret of linear ensemble sampling", *Advances in Neural Information Processing Systems*, vol. 37, pp. 92 803–92 831, 2025.

Experience

Reviewer
NeurIPS 2025 (Top reviewer), ICLR 2026, ICML 2026 (Gold reviewer)

Graduate School of Data Science, Seoul National University — April 2023–May 2025
Undergraduate Researcher
Advisor : Prof. Min-hwan Oh

- Analyzed `FS-WLasso` for sparse linear contextual bandits and derived its $O(\mathrm{poly}\log dT)$ regret bounds
- Improved regret analysis of linear ensemble sampling
- Devised `EQO` for tabular reinforcement learning
- Created algorithmic framework `INFEX` and analyzed regret of infrequent exploration in linear bandits

Thunder Research Group, Dept. of CSE, Seoul National University — Jul 2022–Dec 2022
Undergraduate Research Opportunity Program, Creative Integrated Design 1
Advisor : Prof. Jaejin Lee

- Developed low-precision and quantization methods for GPT-2

Honors and Awards

Paper Awards

K-Data Science Conference — Nov 2024

- President of National Research Foundation of Korea Award (Top 2 among 28 finalists) [5]

Korean Artificial Intelligence Association Summer Conference — Aug 2024

- Excellence Paper Award (Top 7) [5]

Scholarships

KFAS Overseas PhD Scholarship — September 2025-Present
Korea Foundation for Advanced Studies

Merit-Based Scholarship — Spring 2024
Seoul National University

Kwanjeong Domestic Scholarship — Mar 2020–Feb 2024
Kwanjeong Educational Foundation

Eminence Scholarship — Fall 2019
Seoul National University

SNU Development Fund Scholarship — Spring 2019
Seoul National University Foundation

Merit-Based Scholarship — Fall 2018
Seoul National University

Programming Competitions

Silver Medal (7th Place) & Asia Pacific Champion The 47th ICPC World Finals — Apr 2024
4th Place Seoul National University Programming Contest Division 1 — Sep 2023
4th Prize Samsung Collegiate Programming Cup — Sep 2023
Silver Prize (3rd Place) The 2022 ICPC Asia Seoul Regional Contest — Nov 2022
2nd Place The 2022 ICPC Asia Korea National First Round Programming Contest — Oct 2022
4th Place Seoul National University Programming Contest Division 1 — Sep 2022
3rd Prize Samsung Collegiate Programming Cup — Sep 2022

Skills

Programming Languages: **C, C++, Python**
Machine Learning Frameworks: **PyTorch, TensorFlow**
Languages: **Korean** (native), **English** (fluent)